

data_file_eda

Import required library

```
library(dplyr)
library(gtsummary)
library(kableExtra)
library(ggplot2)
```

Global path variable to the data set

```
dataset_path <- read.table(
  "/Users/sorenbasnet/Documents/Github/STAT_443_FINAL_PROJECT/data/data_patient_file_cna_rai/patient_file_cna_rai.csv",
  header = TRUE,
  sep = " "
)
```

```
factor_cols <- c(
  "ONCOTREE_CODE", "DISEASE_TYPE",
  "CANCER_TYPE", "CANCER_TYPE_DETAILED",
  "SOMATIC_STATUS", "CLL_EPITYPE",
  "TUMOR_MOLECULAR_SUBTYPE", "SEQUENCING_PLATFORM",
  "RNA_SEQUENCING", "RRBS", "ARRAY450K",
  "M.CLL_WES", "U.CLL_WES",
  "EC_DISCOVERY", "EC_EXTENSION",
  "IGLV3_21_R110", "U1_STATUS",
  "SEX", "OS_STATUS",
  "COHORT", "IGHV_MUTATION_STATUS",
  "TREATMENT_STATUS", "PRIOR_TREATMENT_CATEGORY",
  "TREATMENT_AFTER_SAMPLING",
  "EXPRESSION_CLUSTER", "FFS_STATUS",
  "X11q_status", "X17p_status", "X17q_status",
  "X18p_status", "X19q_status", "X20p_status",
  "X2p_status", "X4p_status", "X6q_status",
  "X8p_status", "X8q_status", "tri12_status",
  "TREATMENT_STATUS_AT_SAMPLING",
  "FIRST_TREATMENT_AFTER_SAMPLING"
)
```

```
dataset_path[factor_cols] <- lapply(dataset_path[factor_cols], as.factor)
```

```
summary(dataset_path)
```

```
##   SAMPLE_ID      PATIENT_ID      ONCOTREE_CODE DISEASE_TYPE
## Length:1148      Length:1148      CLLSLL:1094  CLL:1094
## Class :character  Class :character  MCBCL : 54   MBL: 54
## Mode :character   Mode :character
##
##
##
```

```

##
## NORMAL_MEAN_COVERAGE TUMOR_MEAN_COVERAGE TUMOR_SAMPLE_PURITY
## Min. : 16.10 Min. : 11.40 Min. :0.3300
## 1st Qu.: 56.30 1st Qu.: 57.25 1st Qu.:0.9100
## Median : 73.10 Median : 75.80 Median :0.9800
## Mean : 77.96 Mean : 81.56 Mean :0.9228
## 3rd Qu.: 88.70 3rd Qu.: 92.95 3rd Qu.:1.0000
## Max. :360.70 Max. :406.90 Max. :1.0000
## NA's :165 NA's :165 NA's :165
## TUMOR_SAMPLE_PLOIDY CANCER_TYPE
## Min. :1.920 Mature__B_Cell__Neoplasms:1148
## 1st Qu.:2.000
## Median :2.000
## Mean :2.005
## 3rd Qu.:2.000
## Max. :2.130
## NA's :165
## CANCER_TYPE_DETAILED
## Chronic__Lymphocytic__Leukemia_Small__Lymphocytic__Lymphoma:1094
## Monoclonal__B_Cell__Lymphocytosis : 54
##
##
##
##
## SOMATIC_STATUS CLL_EPITYPE TUMOR_MOLECULAR_SUBTYPE
## Matched:1148 0 :120 M_CLL:516
## Chemo__+__no__Ab: 1 U_CLL:472
## i_CLL :141 NA's :160
## m_CLL :342
## n_CLL :391
## unclassified : 13
## NA's :140
## SEQUENCING_PLATFORM RNA_SEQUENCING RRBS ARRAY450K M.CL_L_WES U.CL_L_WES
## WES :897 No :438 No :621 No :658 No :636 No :690
## WES_WGS: 86 Yes:710 Yes:527 Yes:490 Yes:512 Yes:458
## WGS : 90
## NA's : 75
##
##
##
## EC_DISCOVERY EC_EXTENSION IGLV3_21_R110 U1_STATUS TMB_NONSYNONYMOUS
## No :547 No :1044 No :1012 MUT : 34 Min. : 0.1333
## Yes:601 Yes: 104 Yes : 106 WT :889 1st Qu.: 0.6000
## NA's: 30 NA's:225 Median : 0.8333
## Mean : 1.0236
## 3rd Qu.: 1.1000
## Max. :18.5667
## NA's :165
## AGE AGE_SAMPLING SEX OS_STATUS OS_MONTHS
## Min. :18.00 Min. :19.00 Female:409 DECEASED:286 Min. : 0.00
## 1st Qu.:51.00 1st Qu.:56.00 Male :735 LIVING :722 1st Qu.: 55.42
## Median :59.00 Median :64.00 NA's : 4 NA's :140 Median : 78.95
## Mean :58.42 Mean :63.39 Mean : 80.15

```

```

## 3rd Qu.:66.00 3rd Qu.:71.00 3rd Qu.:101.64
## Max. :93.00 Max. :94.00 Max. :284.31
## NA's :190 NA's :4 NA's :140
## DEATH_DAYS COHORT IGHV_MUTATION_STATUS IGHV_IDENTITY_PERCENTAGE
## Min. : 0.0 DFCI :179 mutated :581 Min. : 84.72
## 1st Qu.: 0.0 GCLLSG:340 unmutated:543 1st Qu.: 94.10
## Median : 0.0 ICGC :506 NA's : 24 Median : 97.92
## Mean : 493.6 MDACC : 25 Mean : 96.75
## 3rd Qu.: 452.8 NHLBI : 72 3rd Qu.:100.00
## Max. :5723.0 UCSD : 25 Max. :100.00
## NA's :140 NA's : 1 NA's :27
## TREATMENT_STATUS PRIOR_TREATMENT_CATEGORY
## Post_treatment: 97 Ab__+__no__Chemo: 3
## Pre_treatment :1047 Chemo__+__Ab : 37
## NA's : 4 Chemo__+__no__Ab: 26
## LEN__+__Ab : 1
## Rel_Ref : 22
## Untreated :919
## NA's :140
## TREATMENT_AFTER_SAMPLING EXPRESSION_CLUSTER FFS_STATUS
## Chemo__+__Ab :219 0 :441 0:FailureFree:372
## Chemo__+__no__Ab :104 EC_u1 :171 1:Failure :636
## Ab__+__no__chemo : 21 EC_m4 :109 NA's :140
## BTK__inhibitor : 17 EC_u2 : 60
## LEN__+__chemo__+__Ab: 6 EC_i : 59
## (Other) : 29 (Other):168
## NA's :752 NA's :140
## FFS_MONTHS X11q_status X17p_status X17q_status
## Min. : 0.00 Loss : 30 Loss : 71 Gain : 15
## 1st Qu.: 20.18 Unchanged:1032 Unchanged:991 Unchanged:1047
## Median : 46.41 NA's : 86 NA's : 86 NA's : 86
## Mean : 53.53
## 3rd Qu.: 80.71
## Max. :220.30
## NA's :140
## X18p_status X19q_status X20p_status X2p_status
## Loss : 30 Gain : 14 Loss : 16 Gain : 57
## Unchanged:1032 Unchanged:1048 Unchanged:1046 Unchanged:1005
## NA's : 86 NA's : 86 NA's : 86 NA's : 86
##
##
##
## X4p_status X6q_status X8p_status X8q_status
## Loss : 16 Loss : 29 Loss : 31 Gain : 24
## Unchanged:1046 Unchanged:1033 Unchanged:1031 Unchanged:1038
## NA's : 86 NA's : 86 NA's : 86 NA's : 86
##
##
##
## tri12_status RAI_AT_DIAGNOSIS TREATMENT_STATUS_AT_SAMPLING RAI_AT_SAMPLING
## Gain :155 Min. :0.0000 Treated : 67 Min. :0.0000
## Unchanged:907 1st Qu.:0.0000 Untreated:650 1st Qu.:0.0000

```

##	NA's	: 86	Median	:0.0000	NA's	:431	Median	:0.0000
##			Mean	:0.3423			Mean	:0.9679
##			3rd Qu.	:0.0000			3rd Qu.	:2.0000
##			Max.	:4.0000			Max.	:4.0000
##			NA's	:140			NA's	:431
##	FIRST_TREATMENT_AFTER_SAMPLING		surv_5yrs				death_5yr	
##	0	:321	Min.	: 0.00	Min.	:0.0000		
##	Chemo__+__Ab	:219	1st Qu.	:55.42	1st Qu.	:0.0000		
##	Chemo__+__no__Ab	:104	Median	:60.00	Median	:0.0000		
##	Ab__+__no__chemo	: 21	Mean	:52.14	Mean	:0.1627		
##	BTK__inhibitor	: 17	3rd Qu.	:60.00	3rd Qu.	:0.0000		
##	(Other)	: 35	Max.	:60.00	Max.	:1.0000		
##	NA's	:431	NA's	:140	NA's	:140		